CHAITANYA BHARATHI INSTITUTE OF TECHNOLOGY (CBIT)

Department Of Computer Science And Engineering

FUNDAMENTALS OF DATA SCIENCE (22CSC59)



CASE STUDY REPORT

Title: EastWest Airlines Customer Clustering

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1. Problem Statement

East-West Airlines wants to understand its customers based on their flying activity, reward-mile accumulation, credit-card usage and membership history. Customers differ considerably in these behaviours, so treating all customers as one group may limit the effectiveness of loyalty programmes and promotional activities.
The objective of this study is to identify groups of customers with similar characteristics using clustering algorithms. K-Means, Hierarchical Clustering and DBSCAN are applied and compared using silhouette scores, cluster sizes and customer profiles.
The analysis aims to answer the following questions:

1. What major patterns exist in customer travel and reward behaviour?
2. How do the three clustering algorithms group customers?
3. Which tested configuration provides useful customer segments?
4. What actions could the airline consider for different customer groups?

2. Dataset Description

The supplied **EastWestAirlines.xlsx** workbook contains a description sheet and a customer-data sheet. The data sheet contains **3,999 customer records and 12 columns**.
Each row represents one customer. The dataset contains no predefined customer-segment labels, making clustering an unsupervised learning task.

Column	Description
ID#	Unique customer identifier
Balance	Reward miles available for award travel
Qual_miles	Miles counted towards qualifying for Topflight status
cc1_miles	Category of miles earned through the frequent-flyer credit card
cc2_miles	Category of miles earned through the Rewards credit card
cc3_miles	Category of miles earned through the Small Business credit card
Bonus_miles	Miles earned through non-flight bonus transactions
Bonus_trans	Number of non-flight bonus transactions
Flight_miles_12mo	Flight miles earned during the previous 12 months
Flight_trans_12	Number of flight transactions during the previous 12 months
Days_since_enroll	Number of days since enrolment
Award?	Binary indicator of whether the customer received an award during the recorded period

The credit-card mileage columns contain ordered category codes rather than exact mile counts. The ID# column identifies customers but does not describe their behaviour, so it was excluded from clustering.

```
df = pd.read_excel("EastWestAirlines (1).xlsx", sheet_name="data")

print("Dataset shape:", df.shape)
display(df.head())
```

```
Dataset shape: (3999, 12)
```

	ID#	Balance	Qual_miles	cc1_miles	cc2_miles	cc3_miles	Bonus_miles	Bonus_trans	Flight_miles_12mo	Flight_trans_12	Days_since_enroll	Award?
0	1	28143	0	1	1	1	174	1	0	0	7000	0
1	2	19244	0	1	1	1	215	2	0	0	6968	0
2	3	41354	0	1	1	1	4123	4	0	0	7034	0
3	4	14776	0	1	1	1	500	1	0	0	6952	0
4	5	97752	0	4	1	1	43300	26	2077	4	6935	1

3. Methodology

Python was used to perform the analysis. The main libraries were pandas and NumPy for data processing, Matplotlib and Seaborn for visualisation, scikit-learn for clustering and evaluation, and SciPy for the dendrogram.
The workflow consisted of:

1. Loading and inspecting the dataset.
2. Checking missing values, duplicate records and repeated customer IDs.
3. Examining descriptive statistics and visual distributions.
4. Identifying potential outliers using the interquartile range.
5. Applying log transformation and standardisation.
6. Testing K-Means, Hierarchical Clustering and DBSCAN.
7. Examining cluster sizes and average customer characteristics.
8. Comparing results and developing business recommendations.

3.1 Data-quality checks

The dataset contained:

Check	Result
Customer records	3,999
Original columns	12
Missing values	0
Duplicate full rows	0
Repeated customer IDs	0
Features used for clustering	11

No missing-value imputation or duplicate removal was necessary.

3.2 Potential outlier detection

Potential outliers were identified using the IQR rule:
IQR = Q3-Q1 Lower boundary = Q1-1.5(IQR) Upper boundary = Q3+1.5(IQR)
Values outside these boundaries were flagged.

Feature	Lower boundary	Upper boundary	Flagged customers	Percentage
Balance	−92,287.25	203,218.75	266	6.65%
Bonus_miles	−32,575.75	57,626.25	280	7.00%
Flight_miles_12mo	−466.50	777.50	569	14.23%

These are separate feature-level counts. A customer may be flagged in more than one feature, so the counts cannot be added to obtain the number of unique unusual customers.
In the current implementation, flagged customers were retained because unusual values could represent genuine travel or reward behaviour. Their validity was not independently verified.

3.3 Transformation and scaling

Log transformation was applied to:

- Balance
- Qual_miles
- Bonus_miles
- Bonus_trans
- Flight_miles_12mo
- Flight_trans_12
- Days_since_enroll

The transformation used was:
x'=ln(1+x)
This accommodates zero values and compresses large numerical differences. StandardScaler was then applied to the processed features:
z={x'-u}/{sigma}
Standardisation places features on comparable variance scales. It does not remove outliers or guarantee that every feature has equal business importance.
The original customer values were retained separately for interpreting cluster profiles.

4. Exploratory Analysis and Statistical Findings

4.1 Descriptive statistics

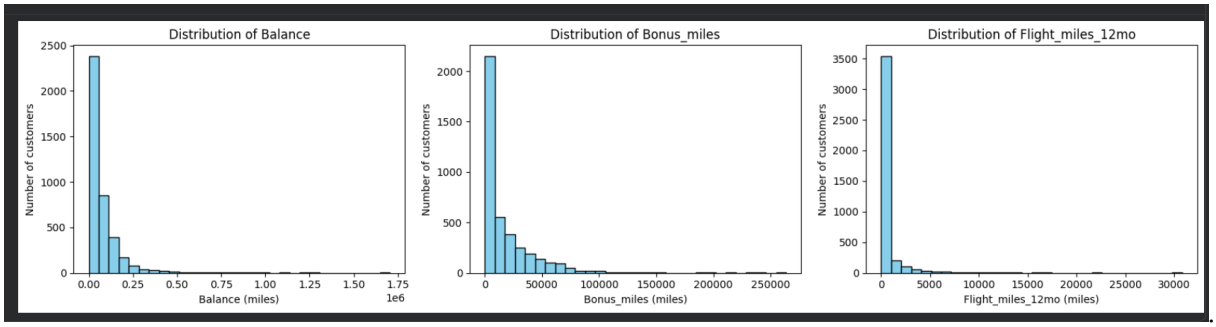
The average Balance was approximately **73,601 miles**, compared with a median of **43,097 miles**.
Average Bonus_miles was approximately **17,145 miles**, compared with a median of **7,171 miles**.
For both features, the mean exceeded the median substantially, consistent with right-skewed distributions.
The 75th percentile of Qual_miles was zero, indicating that at least 75% of customers had no qualifying miles recorded.

```
display(X.describe().round(2))
```

	Balance	Qual_miles	cc1_miles	cc2_miles	cc3_miles	Bonus_miles	Bonus_trans	Flight_miles_12mo	Flight_trans_12	Days_since_enroll	Award?
count	3999.00	3999.00	3999.00	3999.00	3999.00	3999.00	3999.0	3999.00	3999.00	3999.00	3999.00
mean	73601.33	144.11	2.06	1.01	1.01	17144.85	11.6	460.06	1.37	4118.56	0.37
std	100775.66	773.66	1.38	0.15	0.20	24150.97	9.6	1400.21	3.79	2065.13	0.48
min	0.00	0.00	1.00	1.00	1.00	0.00	0.0	0.00	0.00	2.00	0.00
25%	18527.50	0.00	1.00	1.00	1.00	1250.00	3.0	0.00	0.00	2330.00	0.00
50%	43097.00	0.00	1.00	1.00	1.00	7171.00	12.0	0.00	0.00	4096.00	0.00
75%	92404.00	0.00	3.00	1.00	1.00	23800.50	17.0	311.00	1.00	5790.50	1.00
max	1704838.00	11148.00	5.00	3.00	5.00	263685.00	86.0	30817.00	53.00	8296.00	1.00

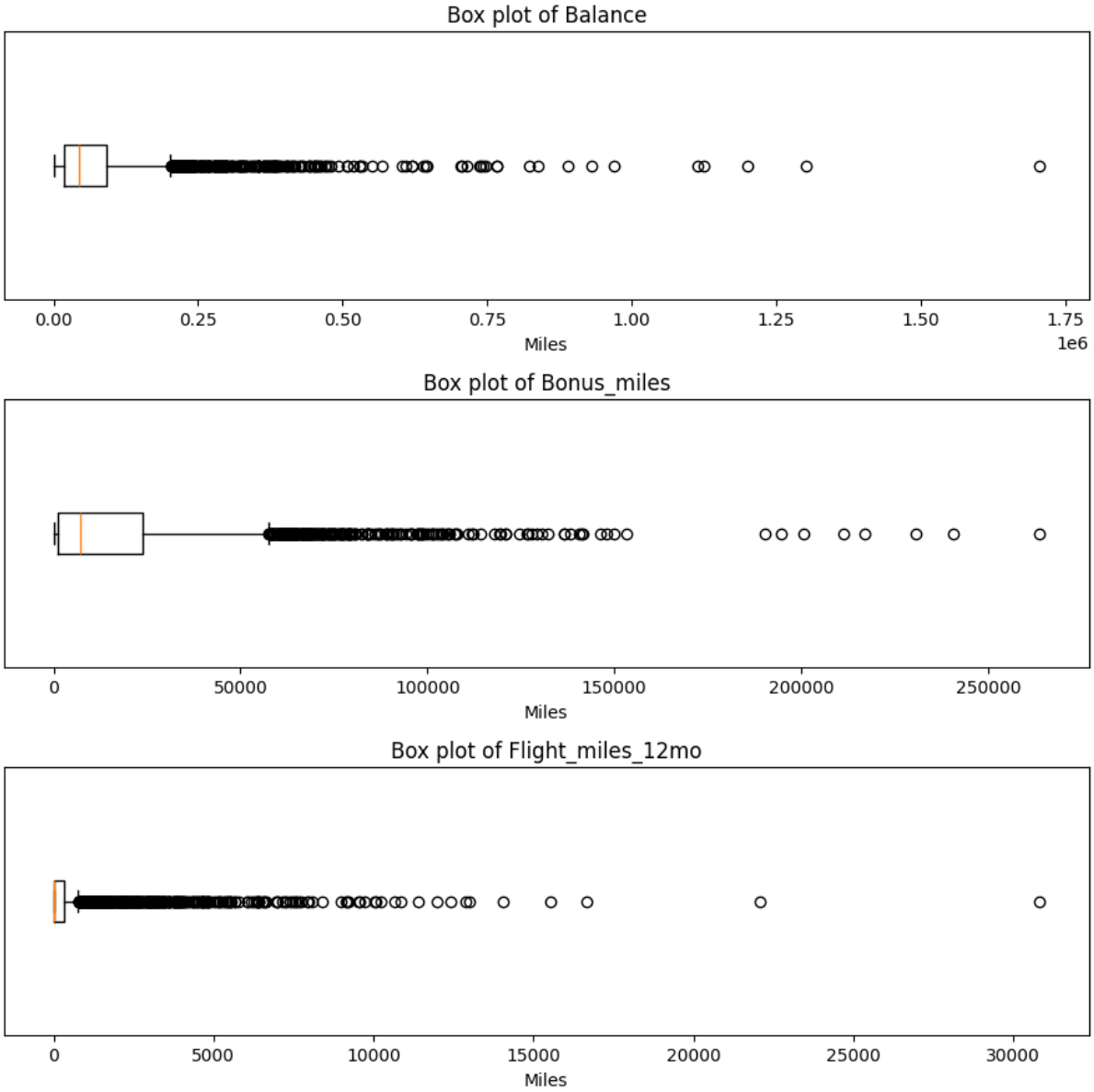
4.2 Histograms

Histograms of Balance, Bonus_miles and Flight_miles_12mo showed that most customers had comparatively low values, while a smaller number had much higher values.
The long right-hand tails illustrate differences in customer activity and support examining transformations before distance-based clustering.



4.3 Box plots

Box plots showed numerous observations beyond the upper whiskers. These observations were treated as potential outliers rather than automatically classified as incorrect records.
The small central box for flight miles indicates that the middle portion of the data is concentrated at low values relative to the largest observations.



4.4 Correlation analysis

A Pearson correlation heatmap was generated for the 11 customer features.
The heatmap was used to examine relationships between flight activity, reward accumulation and other customer characteristics. Strongly related features can give additional weight to similar behaviour in distance calculations.

Correlation does not establish causation. Correlations involving ordered category codes also require cautious interpretation.
Balance
Lower boundary: -92287.25
Upper boundary: 203218.75
Potential outliers: 266
Percentage: 6.65 %

Bonus_miles
Lower boundary: -32575.75
Upper boundary: 57626.25
Potential outliers: 280
Percentage: 7.0 %

Flight_miles_12mo
Lower boundary: -466.5
Upper boundary: 777.5
Potential outliers: 569
Percentage: 14.23 %

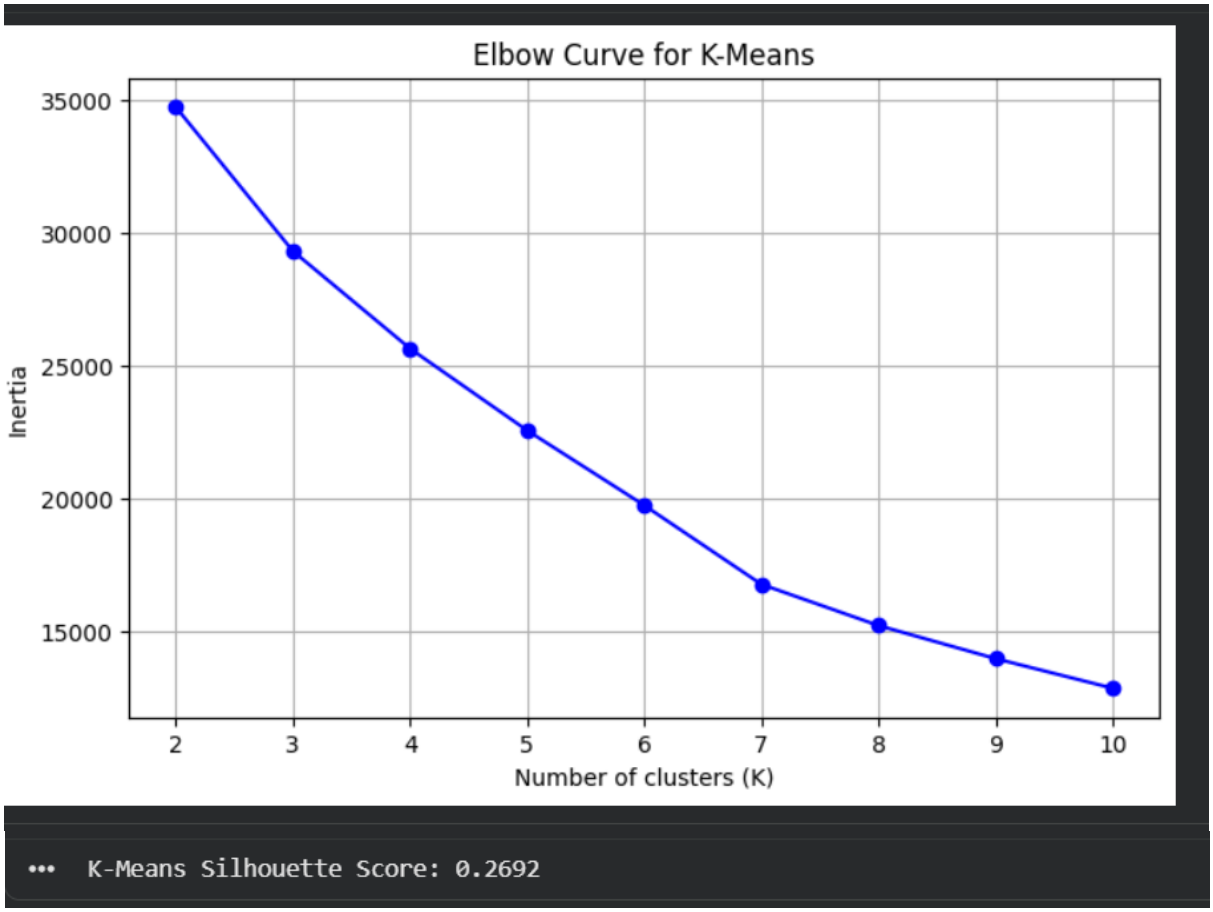
Scaled dataset shape: (3999, 11)

	Balance	Qual_miles	cc1_miles	cc2_miles	cc3_miles	Bonus_miles	Bonus_trans	Flight_miles_12mo	Flight_trans_12	Days_since_enroll	Award?
0	-0.202419	0.0	0.0	0.0	0.0	-0.310281	-0.785714	0.000000	0.0	0.839185	0.0
1	-0.322877	0.0	0.0	0.0	0.0	-0.308463	-0.714286	0.000000	0.0	0.829938	0.0
2	-0.023593	0.0	0.0	0.0	0.0	-0.135163	-0.571429	0.000000	0.0	0.849010	0.0
3	-0.383356	0.0	0.0	0.0	0.0	-0.295825	-0.785714	0.000000	0.0	0.825314	0.0
4	0.739816	0.0	1.5	0.0	0.0	1.602137	1.000000	6.678457	4.0	0.820402	1.0

5. K-Means Clustering

5.1 Parameter evaluation

K-Means groups customers by minimising squared distances to cluster centres. Values of K from 2 to 10 were evaluated using inertia and silhouette scores.
A fixed random state of 42 and 10 initialisations were used for reproducibility.
Inertia measures within-cluster squared distances and generally decreases as the number of clusters increases. The elbow curve helps examine diminishing improvements but does not always identify one clear cluster count.



Five clusters were retained as a practical exploratory configuration. This choice was not the highest-scoring configuration across the entire tested range and should not be interpreted as a proven optimum.

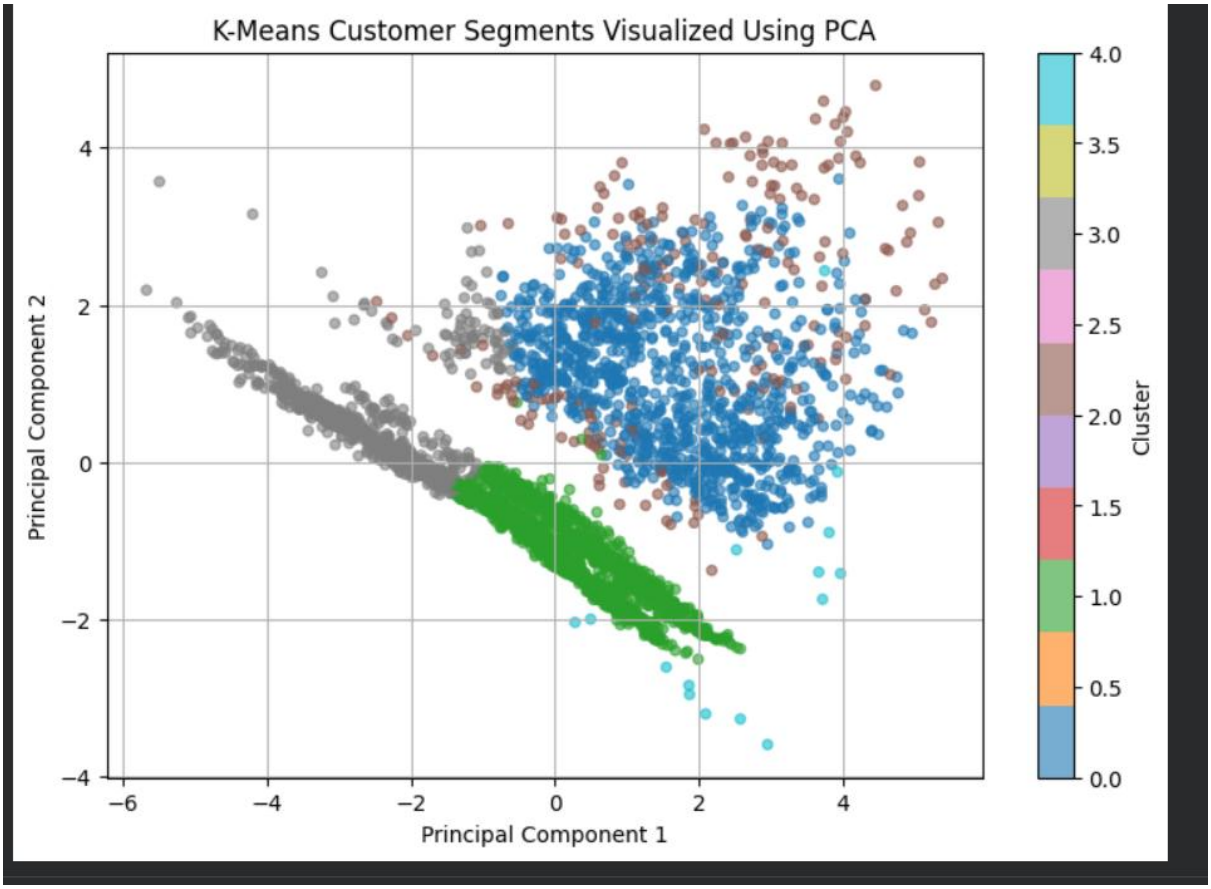
5.2 Results and interpretation

The five-cluster model achieved a silhouette score of **0.2692**, indicating limited separation and considerable overlap between customer groups.

Cluster	Customers	Interpretation
0	1,048	Active travellers with substantial rewards
1	1,716	Reward accumulation with almost no recent flight activity
2	214	Higher qualifying miles and greater flight activity
3	1,006	Low engagement in travel and reward accumulation
4	15	Small group with high bonus miles and higher business-card categories

Cluster 0 averaged approximately 1,358 flight miles and 4.02 flight transactions.
Cluster 1 averaged approximately 20,237 bonus miles but almost no flight activity, suggesting reward accumulation mainly through non-flight activities.
Cluster 2 had the highest average flight activity, approximately 1,806 miles and 5.44 transactions, and higher qualifying miles.
Cluster 3 had the lowest average balance and bonus-mile accumulation, together with minimal flight activity.
Cluster 4 averaged approximately 93,928 bonus miles and had higher small-business credit-card categories. Its small size requires cautious interpretation.

Cluster	Balance	Bonus_miles	Flight_miles_12mo	Flight_trans_12	Days_since_enroll	Award?
0	108948.58	24996.10	1358.07	4.02	4473.27	0.57
1	73704.73	20236.73	0.09	0.00	4355.68	0.37
2	139046.61	25291.85	1805.76	5.44	4418.90	0.68
3	21719.08	813.84	22.18	0.09	3273.30	0.08
4	138061.40	93927.87	506.67	1.60	4613.87	0.53



PCA was used only for visualisation. Clustering was performed using all 11 processed features.

6. Hierarchical Clustering

Agglomerative clustering initially treats each customer as a separate group and progressively merges groups.

Ward, complete and average linkage were compared while requesting five clusters.

Linkage method	Silhouette score
Ward	0.2634
Complete	0.2709
Average	0.5582

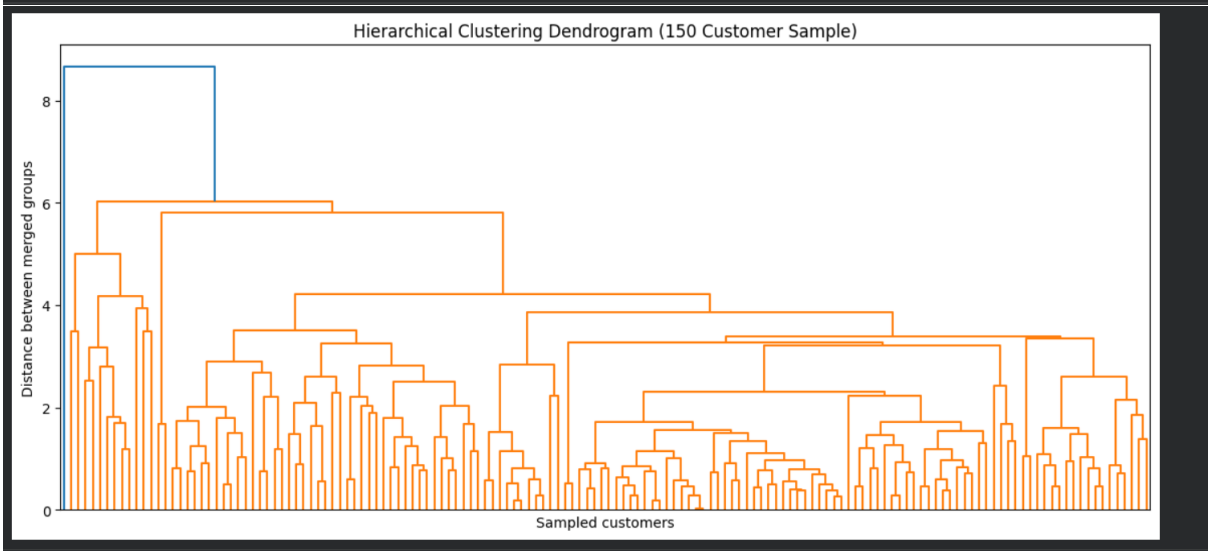
Average linkage had the highest score and was examined further.

Cluster	Customers	Percentage
0	3,942	98.57%
1	43	1.08%
2	2	0.05%
3	11	0.28%
4	1	0.03%

Percentages are rounded.
Although the score was high, almost all customers were placed in one group. Small groups included customers with higher credit-card mileage categories, two very recently enrolled customers and one unusual customer.
This result demonstrates that a high silhouette score does not necessarily produce useful broad customer segmentation.

```
Linkage: ward     Silhouette Score: 0.2634
Linkage: complete Silhouette Score: 0.2709
Linkage: average  Silhouette Score: 0.5582
```





The dendrogram illustrates the merging process on a sample. The final hierarchical model was fitted to the full processed dataset.

7. DBSCAN

DBSCAN identifies dense groups using two parameters:

- **eps:** maximum neighbourhood distance.
- **min_samples:** minimum number of observations, including the point itself, required for a core point.

Customers that do not belong to a detected cluster receive the label −1 and are treated as noise.

7.1 Parameter evaluation

Multiple neighbourhood distances and minimum-sample values were tested.

	eps	Clusters	Noise points	Silhouette score
0	0.8	10	813	0.2668
1	1.0	6	451	0.2643
2	1.2	10	269	0.2616
3	1.5	11	135	0.1757
4	2.0	6	59	0.2097
5	2.2	3	53	0.3899
6	2.5	4	30	0.3919
7	3.0	4	23	0.3918

Across these settings, approximately 92.8% of customers remained in the largest cluster. Changes mainly affected smaller groups and noise assignments.
The highest score in the additional trials was **0.3923** at eps=3.0 and min_samples=5.
The selected configuration, eps=2.5 and min_samples=10, achieved **0.3919**. It was retained as a configuration with nearly the same score and a stricter minimum neighbourhood size.

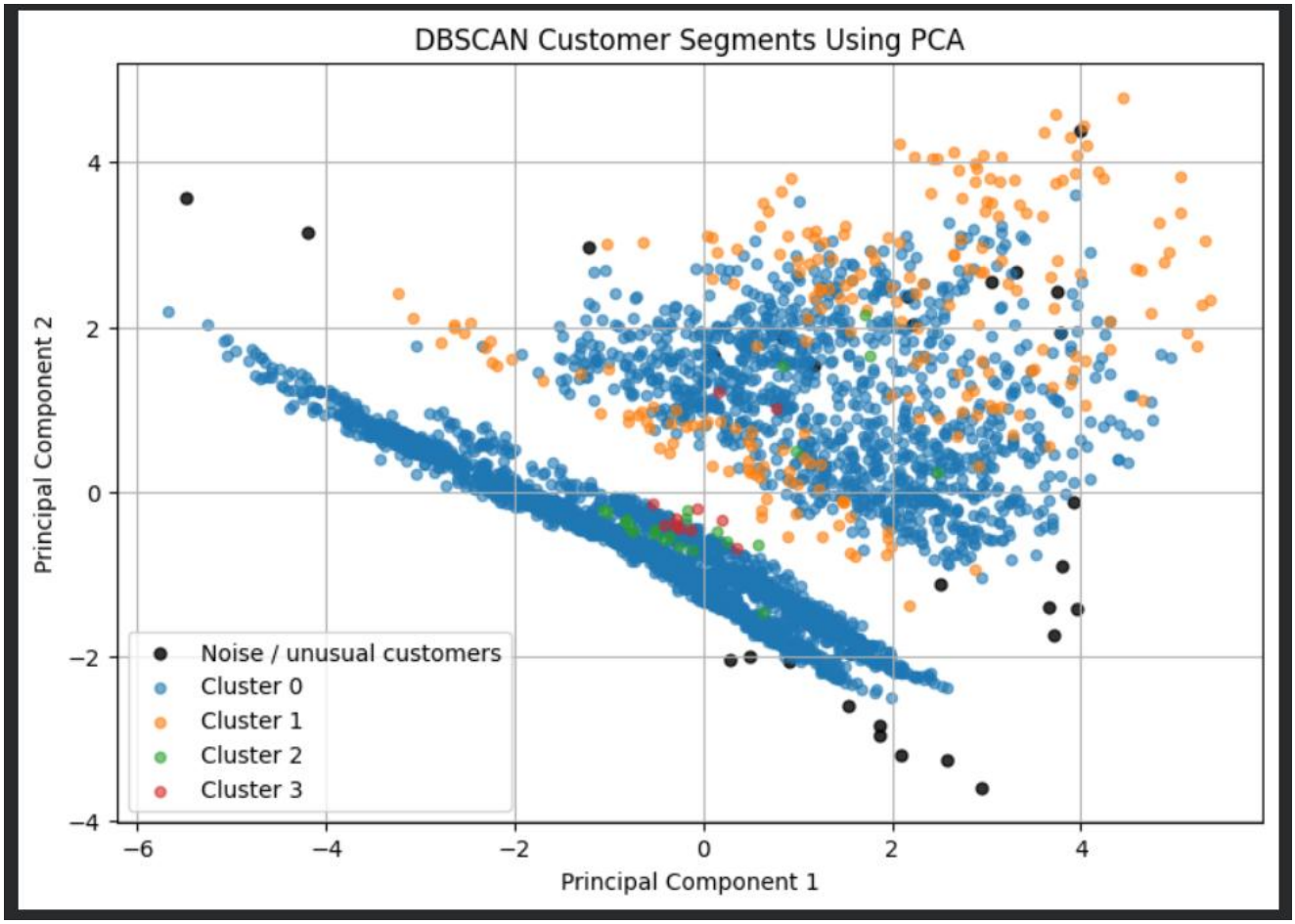
7.2 Results and interpretation

The selected DBSCAN configuration identified four clusters and 30 noise customers.

Group	Customers	Percentage
Cluster 0	3,713	92.85%
Cluster 1	222	5.55%
Cluster 2	23	0.58%
Cluster 3	11	0.28%
Noise	30	0.75%

Cluster 0 represented the large main customer group with zero qualifying miles.
Cluster 1 had higher qualifying miles and greater flight activity.
Clusters 2 and 3 were small groups distinguished by Rewards credit-card mileage categories.
Noise customers had unusual combinations of features. Noise labels do not prove that their records are errors.

```
Number of DBSCAN clusters: 4
Noise points: 30
DBSCAN Silhouette Score: 0.3919
```



8. Model Evaluation and Comparison

Algorithm	Clusters	Noise customers	Silhouette score
K-Means	5	0	0.2692
Hierarchical—average linkage	5	0	0.5582
DBSCAN	4	30	0.3919

Silhouette scores measure within-cluster similarity relative to separation from other clusters. Average-linkage hierarchical clustering achieved the highest score, but 98.57% of customers belonged to one cluster. DBSCAN also placed 92.85% of customers in one group.

K-Means produced several substantial groups with different behaviours, making it more useful for broad segmentation among the selected configurations. Its lower score shows that these behavioural groups overlap.

DBSCAN's score was calculated after excluding noise customers. K-Means and hierarchical scores used all customers, so the scores refer to slightly different evaluation populations.

9. Business Recommendations

Customer group	Suggested action
Active reward travellers	Test flight-related bonus-mile promotions
Reward earners with little recent flight activity	Encourage travel through reward-redemption offers
Customers with higher flight activity	Test loyalty and retention benefits
Low-engagement customers	Use simple reactivation campaigns
Small business-card segment	Review individual patterns before launching specialised offers

Customer group	Suggested action
Unusual customers	Investigate their behaviour and data quality individually

These recommendations are based on observed patterns. Their effectiveness would need to be tested through subsequent campaigns.

10. Limitations

1. Clusters depend on feature selection, transformation, scaling and parameter settings.
2. Numerical measurements, ordered category codes and a binary award indicator were combined in distance calculations.
3. Potential outliers were retained; their validity was not independently confirmed.
4. Small clusters may be sensitive to individual customers or modelling choices.
5. PCA plots show only part of the information in the original feature space.
6. There are no known correct segment labels against which to validate the clusters.
7. Silhouette scores do not measure customer profitability or campaign effectiveness.

11. Conclusion

This study applied K-Means, Hierarchical Clustering and DBSCAN to airline customer data. K-Means provided the most useful broad segmentation among the selected configurations, distinguishing customers by flight activity, reward accumulation and engagement. Average-linkage hierarchical clustering achieved the highest silhouette score but placed almost all customers in one group. DBSCAN identified smaller dense groups and unusual customers while also retaining a dominant main cluster.

The results demonstrate that selecting a clustering solution requires examining customer profiles, cluster sizes and practical usefulness alongside numerical evaluation metrics.

12. References and Acknowledgement

1. Faculty-provided **Clustering Analysis** assignment.
2. Faculty-provided **EastWestAirlines.xlsx** dataset and its Description sheet.
3. pandas documentation.
4. NumPy documentation.
5. scikit-learn documentation for preprocessing, clustering, PCA and silhouette evaluation.
6. Matplotlib, Seaborn and SciPy documentation.
7. ChatGPT assistance for explanations, code guidance, debugging and report organisation. The notebook was executed and its outputs checked by the student.